

Supplementary Document for “Complexity-Reduced Algorithms for Network and Network-Temporal Kernel Density Visualization”

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I How to Support Other Kernels?

In the paper, we mainly focus on developing complexity-reduced algorithms for generating NKDV and NTKDV using the Epanechnikov function as the spatial and temporal kernels. Here, we further investigate how to extend our NKDV and NTKDV methods for supporting other kernel functions (i.e., uniform and quartic kernels in Table 2) in Sections A and B, respectively.

A NKDV with Other Kernels

Recall from the core idea 2 that we can adopt the ADA method (see Section 3.1) and the ADA^{B+} method (cf. Section 3.2) for supporting other kernel functions as long as their edge- e network kernel density functions $f_e(q)$ (with other kernels) can be represented by the aggregate distance terms (see Equations 11 and 12). Here, we show that both the uniform and quartic kernels fulfill this condition.

Uniform kernel. Consider the edge- e network kernel density function $f_e(q)$ with the uniform kernel, i.e.,

$$f_e(q) = \sum_{p \in P(e)} w \cdot \begin{cases} \frac{1}{b_G} & \text{if } d_G(q, p) \leq b_G \\ 0 & \text{otherwise} \end{cases}$$

Based on Figure 10, we have

$$\begin{aligned} f_e(q) &= \sum_{p \in P(u, p_r)} \frac{w}{b_G} + \sum_{p \in P(v, p_l)} \frac{w}{b_G} \\ &= \frac{w}{b_G} a_{P(u, p_r)}^{(0)} + \frac{w}{b_G} a_{P(v, p_l)}^{(0)} \end{aligned}$$

which can be represented by the aggregate distance terms (see Equations 11 and 12) with $deg = 0$.

Quartic kernel. Consider the edge- e network kernel density function $f_e(q)$ with the

quartic kernel, i.e.,

$$f_e(q) = \sum_{p \in P(e)} w \cdot \begin{cases} (1 - \frac{1}{b_G^2} d_G(q, p))^2 & \text{if } d_G(q, p) \leq b_G \\ 0 & \text{otherwise} \end{cases}$$

Based on Figure 10, we have

$$\begin{aligned} f_e(q) &= \sum_{p \in P(u, p_r)} w \cdot \left(1 - \frac{1}{b_G^2} (d_G(q, u) + d_G(u, p))^2\right)^2 \\ &\quad + \sum_{p \in P(v, p_l)} w \cdot \left(1 - \frac{1}{b_G^2} (d_G(q, v) + d_G(v, p))^2\right)^2 \\ &= w \cdot \left(1 - \frac{1}{b_G^2} d_G(q, u)^2\right)^2 a_{P(u, p_r)}^{(0)} - \frac{4w \cdot d_G(q, u)}{b_G^2} \left(1 - \frac{1}{b_G^2} d_G(q, u)^2\right) a_{P(u, p_r)}^{(1)} \\ &\quad - \frac{2w}{b_G^2} \left(1 - \frac{3}{b_G^2} d_G(q, u)^2\right) a_{P(u, p_r)}^{(2)} + \frac{4w \cdot d_G(q, u)}{b_G^4} a_{P(u, p_r)}^{(3)} + \frac{w}{b_G^4} a_{P(u, p_r)}^{(4)} \\ &\quad + w \cdot \left(1 - \frac{1}{b_G^2} d_G(q, v)^2\right)^2 a_{P(v, p_l)}^{(0)} - \frac{4w \cdot d_G(q, v)}{b_G^2} \left(1 - \frac{1}{b_G^2} d_G(q, v)^2\right) a_{P(v, p_l)}^{(1)} \\ &\quad - \frac{2w}{b_G^2} \left(1 - \frac{3}{b_G^2} d_G(q, v)^2\right) a_{P(v, p_l)}^{(2)} + \frac{4w \cdot d_G(q, v)}{b_G^4} a_{P(v, p_l)}^{(3)} + \frac{w}{b_G^4} a_{P(v, p_l)}^{(4)} \end{aligned}$$

which can be represented by the aggregate distance terms (see Equations 11 and 12) with $deg = 0, 1, 2, 3, 4$.

B NTKDV with Other Kernels

Recall from Section 4.1 that all methods, $\text{ADA}_{\text{TIC}}^{\text{B}+}$ and $\text{ADA}_{\text{TIS}}^{\text{B}+}$, can be extended to support other kernel functions if the network-temporal kernel density function $\mathcal{F}_P(q, t_i)$ can be decomposed into multiple $\sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot t_p^g \cdot K_G(q, p)$ terms with different positive integers g (like Equation 20). Here, we show that both the uniform and quartic kernels can satisfy this condition.

Uniform kernel. Consider the uniform function as a temporal kernel in $\mathcal{F}_P(q, t_i)$ and a set $\text{TIC}(t_i)$. We have

$$\mathcal{F}_P(q, t_i) = \frac{1}{b_T} \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot K_G(q, p)$$

Hence, $\mathcal{F}_P(q, t_i)$ can be decomposed into $\sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot t_p^g \cdot K_G(q, p)$ with $g = 0$.

Quartic kernel. Consider the quartic function as a temporal kernel in $\mathcal{F}_P(q, t_i)$ and a

set $\text{TIC}(t_i)$. We have

$$\begin{aligned}
& \mathcal{F}_P(q, t_i) \\
&= \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot K_G(q, p) \cdot \left(1 - \frac{1}{b_T^2} d(t_i, t_p)^2\right)^2 \\
&= \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot K_G(q, p) \cdot \left(1 - \frac{1}{b_T^2} (t_i^2 - 2t_i t_p + t_p^2)\right)^2 \\
&= \left(1 - \frac{t_i^2}{b_T^2}\right)^2 \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot K_G(q, p) + \frac{4t_i}{b_T^2} \left(1 - \frac{t_i^2}{b_T^2}\right) \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot t_p \cdot K_G(q, p) \\
&\quad - \frac{2}{b_T^2} \left(1 - \frac{3t_i^2}{b_T^2}\right) \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot t_p^2 \cdot K_G(q, p) - \frac{4t_i}{b_T^4} \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot t_p^3 \cdot K_G(q, p) \\
&\quad + \frac{1}{b_T^4} \sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot t_p^4 \cdot K_G(q, p)
\end{aligned}$$

Hence, $\mathcal{F}_P(q, t_i)$ can be decomposed into $\sum_{(p, t_p) \in \text{TIC}(t_i)} w \cdot t_p^g \cdot K_G(q, p)$ with $g = 0, 1, 2, 3, 4$.

II Experiments with Other Kernels

In the paper, we only test the efficiency for generating NKDV and NTKDV using the default Epanechnikov function as the spatial and temporal kernels. Here, we proceed to investigate the efficiency of all NKDV and NTKDV methods with other kernel functions, including uniform and quartic kernels. As a remark, we adopt two datasets, Detroit and San Francisco, for conducting the experiments.

Varying the spatial bandwidth b_G (for NKDV). To conduct this experiment, we choose five values of b_G , 500m, 1000m, 2000m, 5000m, and 10000m, for testing the response time of each NKDV method. Figure I shows the results of all methods. Note that the trends of response time (e.g., Figure Ia and Figure Ic) are similar to the trends using the Epanechnikov kernel (e.g., Figure 28b) no matter which kernel function we adopt. Since our methods, ADA and ADA^{B+}, have the lower time complexities for generating NKDV (regardless of kernel functions), there are visible time gaps between ADA (or ADA^{B+}) and the existing methods, RQS and SPS. Our best method, ADA^{B+}, can achieve 4.85x to 46.36x speedup compared with the state-of-the-art SPS method.

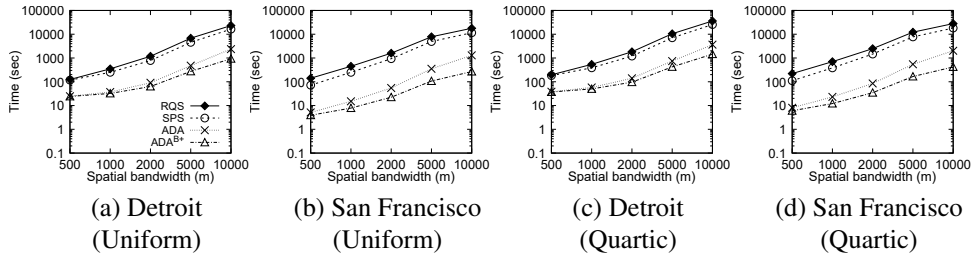


Figure I: Response time for generating NKDV in the Detroit ((a) and (c)) and the San Francisco ((b) and (d)) datasets with the uniform ((a) and (b)) and the quartic ((c) and (d)) kernels, varying the spatial bandwidth b_G .

Varying the temporal bandwidth b_T (for NTKDV). We proceed to examine how b_T can affect the response time for generating NTKDV using other kernel functions. To conduct this experiment, we vary b_T from 7 days to 35 days and measure the response time of each method. Figure II shows the results of all methods. Observe that the trends of response time of all methods in each dataset (e.g., Figure IIa and Figure IIc) are similar to the ones using the Epanechnikov kernel (e.g., Figure 31b) no matter which kernel function we choose. In addition, like the explanation for the results in Figure 31, the response time of each method is also not sensitive to b_T . Furthermore, due to the lower time complexities of our methods, ADA_{TIC}^{B+} and ADA_{TIS}^{B+} outperform the existing methods, RQS and SPS, by 11.45x to 77.09x.

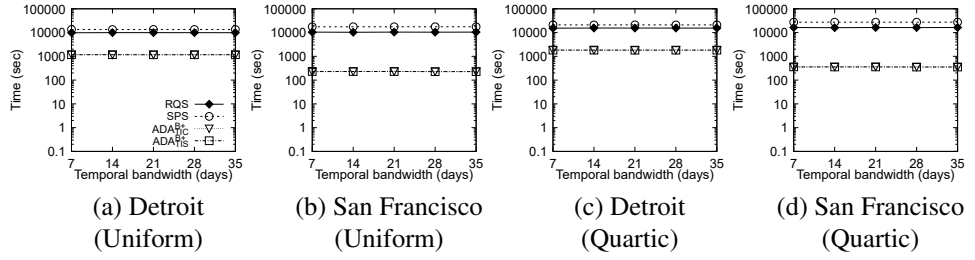


Figure II: Response time for generating NTKDV in the Detroit ((a) and (c)) and the San Francisco ((b) and (d)) datasets with the uniform ((a) and (b)) and the quartic ((c) and (d)) kernels, varying the temporal bandwidth b_T .